

Customer Shopping Trends Indian: Consumer Behavior Analysis

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Chapter 1

Introduction

Dataset used in this report: https://www.kaggle.com/datasets/rewantbhriguvanshi/customer-shopping-trends-indian?select=customer_shopping_behavior-clean.csv

1.1 About the Dataset

- The Customer Shopping Trends Indian Dataset offers insights into consumer behavior and purchasing patterns. Understanding customer preferences and trends is critical for businesses to tailor their products, marketing strategies, and overall customer experience. This dataset captures a wide range of customer attributes including age, gender, purchase history, preferred payment methods, frequency of purchases, all in an Indian context. Analyzing this data can help businesses make informed decisions, optimize product offerings, and enhance customer satisfaction. It is important to note that this dataset is a **synthetic dataset created for beginners** to learn more about data analysis.
- This dataset was built using a custom Python relational engine via Gemini designed to mimic the actual physical and logistical constraints of Indian Retail & E-Commerce.

1.2 Key Features

- **Brand & Category Logic:** Brands only sell what they actually make (e.g., Nike strictly sells activewear/footwear; Biba strictly sells traditional wear).

- **Platform Exclusivity:** Flagship sales are locked to their real-world platforms and months (e.g., Big Billion Days only occurs on Flipkart in Sept/Oct).
- **Economic Tiers:** Pricing uses a Base Cost $\times$ Brand Premium Multiplier logic, ensuring luxury brands cost more and large items cost more than accessories.
- **Logistical Integrity:** "Same Day" delivery always equals 0 days. In-store purchases correctly have no delivery time, and premium brands generally yield higher review ratings.

1.3 Dataset Glossary (Column-Wise)

- Transaction ID – Unique identifier for each purchase.
- Customer ID – Unique identifier for each customer.
- Purchase Date – Date of the transaction (Jan 2023 – Dec 2024).
- Age – Customer's age.
- Gender – Customer's gender.
- Location – Customer's city (major Indian cities).
- Purchase Channel (Online/Offline).
- Online Store – Platform used (Amazon, Flipkart, Myntra).
- Category – Broad product category (Electronics, Clothing, Grocery).
- Item – Specific product purchased.
- Brand – Brand of the purchased item.
- Color – Product color.
- Size – Product size (Clothing: XS–XXL; Footwear: 6–11).
- Quantity – Number of units purchased.
- Purchase Amount () – Final transaction value.
- Discount (%) – Percentage discount applied.

- Festival/Sale Event – Promotional event or Standard Day.
- Shipping Charge () – Delivery fee.
- Delivery Speed/Time – Delivery timeline.
- Subscription Status – Premium membership status.
- Payment Method – Mode of payment (UPI, Credit Card, etc.).
- Review Rating – Customer rating (1–5 stars).
- Return Status – Indicates whether the item was returned.
- Frequency of Purchases – Shopping frequency.
- Previous Purchases – Total number of historical purchases.

Chapter 2

SQL Coding

Listing 2.1: Python + DuckDB code for data analysis

```
import duckdb
import pandas as pd

csv_file = "../customer_shopping_behavior-clean.csv" #
    adjust path if needed

# 1. Analyze best-selling products by gender
query1 = f"""
SELECT Gender, "Item Purchased", COUNT(*) AS total_orders
FROM read_csv_auto('{csv_file}')
GROUP BY Gender, "Item Purchased"
ORDER BY total_orders DESC;
"""

# 2. Return rate by payment method
query2 = f"""
SELECT "Payment Method",
        COUNT(CASE WHEN "Return Status" = 'Returned' THEN 1
        END) AS returned_orders,
        COUNT(CASE WHEN "Return Status" = 'Not Returned'
        THEN 1 END) AS successful_orders,
        ROUND(100.0 * COUNT(CASE WHEN "Return Status" = '
        Returned' THEN 1 END) / COUNT(*), 2) AS return_rate
FROM read_csv_auto('{csv_file}')
GROUP BY "Payment Method";
"""

# 3. Revenue by age group
```

```

query3 = f"""
SELECT
    CASE
        WHEN Age < 25 THEN '<25'
        WHEN Age BETWEEN 25 AND 35 THEN '25-35'
        WHEN Age BETWEEN 36 AND 45 THEN '36-45'
        ELSE '>46'
    END AS AgeGroup,
    SUM("Purchase Amount ()") AS revenue
FROM read_csv_auto('{csv_file}')
GROUP BY AgeGroup
ORDER BY revenue DESC;
"""

# 4. Repeat purchase frequency by customer
query4 = f"""
SELECT "Customer ID", COUNT(*) AS purchase_count
FROM read_csv_auto('{csv_file}')
GROUP BY "Customer ID"
HAVING COUNT(*) > 1
ORDER BY purchase_count DESC;
"""

# 5. Most popular payment method by age group
query5 = f"""
WITH PaymentStats AS (
    SELECT
        CASE
            WHEN Age < 25 THEN '<25'
            WHEN Age BETWEEN 25 AND 35 THEN '25-35'
            WHEN Age BETWEEN 36 AND 45 THEN '36-45'
            ELSE '>46'
        END AS AgeGroup,
        "Payment Method",
        COUNT(*) AS total_orders,
        ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (
            PARTITION BY
                CASE
                    WHEN Age < 25 THEN '<25'
                    WHEN Age BETWEEN 25 AND 35 THEN '25-35'
                    WHEN Age BETWEEN 36 AND 45 THEN '36-45'
                    ELSE '>46'
                END
        )) AS percentage
    FROM read_csv_auto('{csv_file}')
)

```

```

            ), 2) AS percentage
        FROM read_csv_auto('{csv_file}')
        GROUP BY AgeGroup, "Payment Method"
    )
    SELECT AgeGroup, "Payment Method", total_orders, percentage
    FROM (
        SELECT *,
            ROW_NUMBER() OVER (PARTITION BY AgeGroup ORDER BY
                total_orders DESC) AS rn
        FROM PaymentStats
    ) t
    WHERE rn = 1
    ORDER BY AgeGroup;
"""

# 6. Average review rating by delivery speed
query6 = f"""
SELECT "Delivery Speed", AVG("Review Rating") AS avg_rating
,
COUNT(*) AS total_orders
FROM read_csv_auto('{csv_file}')
GROUP BY "Delivery Speed"
ORDER BY avg_rating DESC;
"""

```


Chapter 3

Results

3.1 Best-Selling Products by Gender

Gender	Item Purchased	Total Orders
Male	Kurta	435
Male	Running Shoes	426
Female	Saree	418
Female	Sneakers	408
Male	Saree	407

Table 3.1: Top products purchased by gender

3.2 Return Rate by Payment Method

Payment Method	Returned Orders	Successful Orders	Return Rate (%)
Debit Card	260	1160	18.31
Cash	99	436	18.50
Wallet	54	291	15.65
Cash on Delivery	183	908	16.77
Credit Card	344	1631	17.42

Table 3.2: Return rate comparison across payment methods

3.3 Revenue by Age Group

Age Group	Revenue ()
>46	26,192,400
25–35	14,894,338
36–45	14,353,350
<25	9,236,857

Table 3.3: Revenue distribution across age groups

3.4 Repeat Customers

Customer ID	Purchase Count
CUST002895	10
CUST000316	10
CUST001753	10
CUST001208	9
CUST000449	9

Table 3.4: Top customers with repeat purchases

3.5 Most Popular Payment Method by Age Group

Age Group	Payment Method	Total Orders	Percentage (%)
25–35	UPI	999	43.45
36–45	UPI	921	42.27
<25	UPI	603	42.17
>46	UPI	1756	42.91

Table 3.5: Dominant payment method by age group

3.6 Average Review Rating by Delivery Speed

Delivery Speed	Avg Rating	Total Orders
Express	3.67	2167
N/A (Offline)	3.66	2583
Same Day	3.66	729
Standard	3.26	4521

Table 3.6: Impact of delivery speed on average review rating

Chapter 4

Insights for E-Commerce Application Development

Based on the analysis results, the following insights can be leveraged to design and optimize an e-commerce application:

Product Recommendations

- Male customers frequently purchase **Kurta** and **Running Shoes**, while female customers prefer **Saree** and **Sneakers**.
- Traditional clothing items (Kurta, Saree) remain popular across genders.
- **App Feature:** Implement personalized product recommendations by gender and past purchase history.

Return Behavior and Payment Methods

- **Cash** and **Debit Card** transactions show the highest return rates (18.5%).
- **Wallet** payments have the lowest return rate (15.65%), suggesting higher trust and satisfaction.
- **App Feature:** Highlight secure digital payment options (UPI, Wallet) in checkout flow to reduce returns.
- **App Feature:** Add clear return policy prompts for Cash and Debit Card users to build confidence.

Revenue by Age Group

- Customers aged **over 46** generate the highest revenue (26.2M).
- Middle-aged groups (25–35 and 36–45) also contribute significantly.
- Younger customers (<25) spend less overall.
- **App Feature:** Segment marketing campaigns by age group, offering premium products to older users and budget-friendly options to younger users.

Customer Loyalty

- Several customers have made **10 repeat purchases**, showing strong loyalty.
- **App Feature:** Introduce a loyalty program with badges, discounts, or exclusive offers for repeat buyers.
- **App Feature:** Personalized dashboards showing purchase milestones to encourage continued engagement.

Payment Preferences by Age Group

- Across all age groups, **UPI** is the dominant payment method (over 42% of transactions).
- **App Feature:** Default checkout option should prioritize UPI, while still offering flexibility for other methods.
- **App Feature:** Provide age-specific payment suggestions (e.g., highlight UPI for younger users, Credit Card for older users).

Delivery Speed and Customer Satisfaction

- Faster delivery (**Express, Same Day**) yields higher ratings (3.67).
- **Standard** delivery has the lowest rating (3.26).
- Offline purchases maintain strong ratings (3.66).

- **App Feature:** Offer premium delivery options (Express, Same Day) as upsell features.
- **App Feature:** Display estimated delivery times clearly to manage expectations and improve satisfaction.

Strategic Implications

- Personalize product recommendations by gender and age.
- Optimize checkout flow to highlight UPI and Wallet payments.
- Develop loyalty programs to retain high-frequency customers.
- Segment marketing campaigns by age group revenue contribution.
- Invest in faster delivery options to improve ratings and customer trust.